

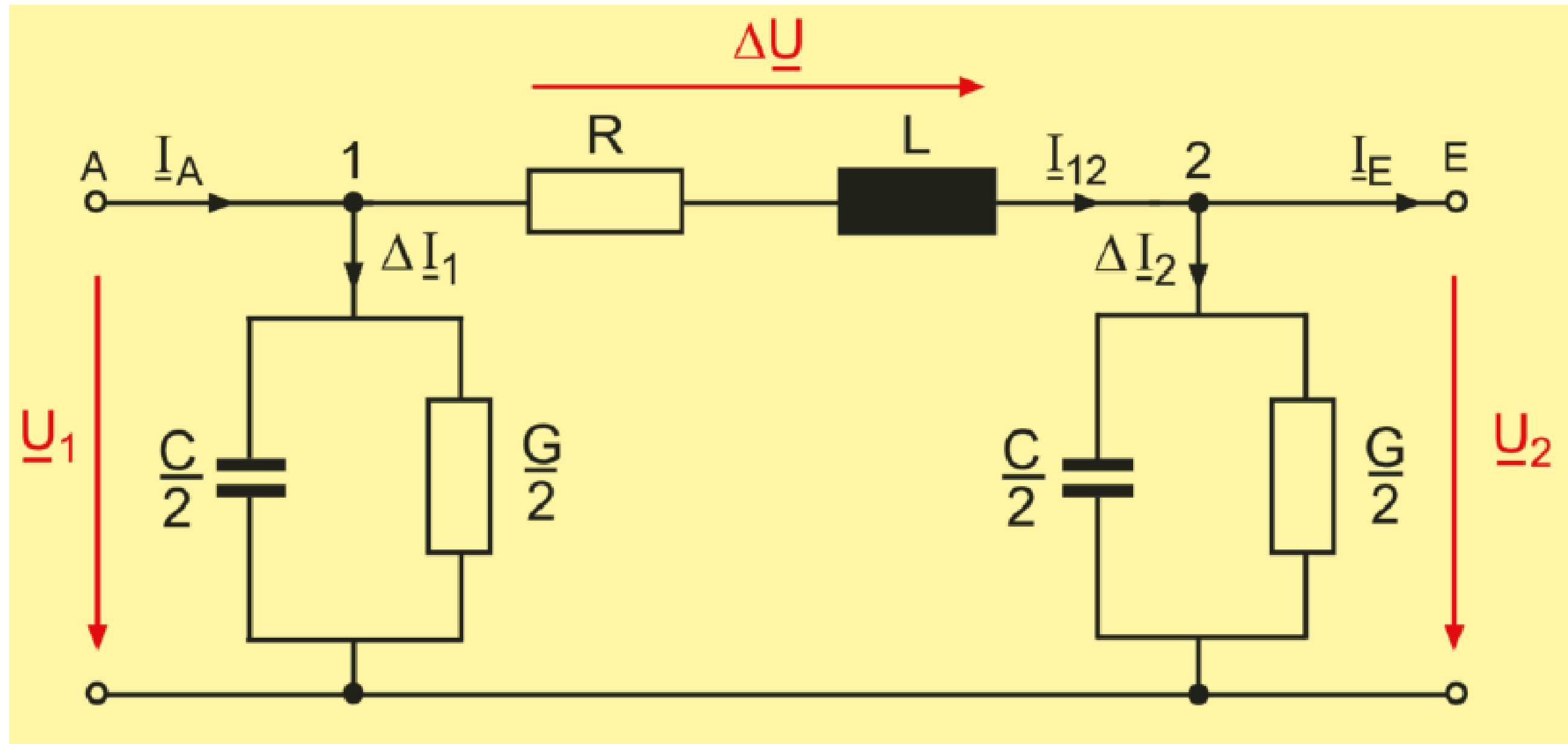
Compensation Units – Energy Storage

Andreas Nenning
Winter Term 2026
Course Number 074742056202

Control of power flow - Compensation Unites

- ♦ Traditionally unidirectional and static
- ♦ Control of power flow
 - traded power volume increased
 - integration of distributed energy – power flow **more dynamic**

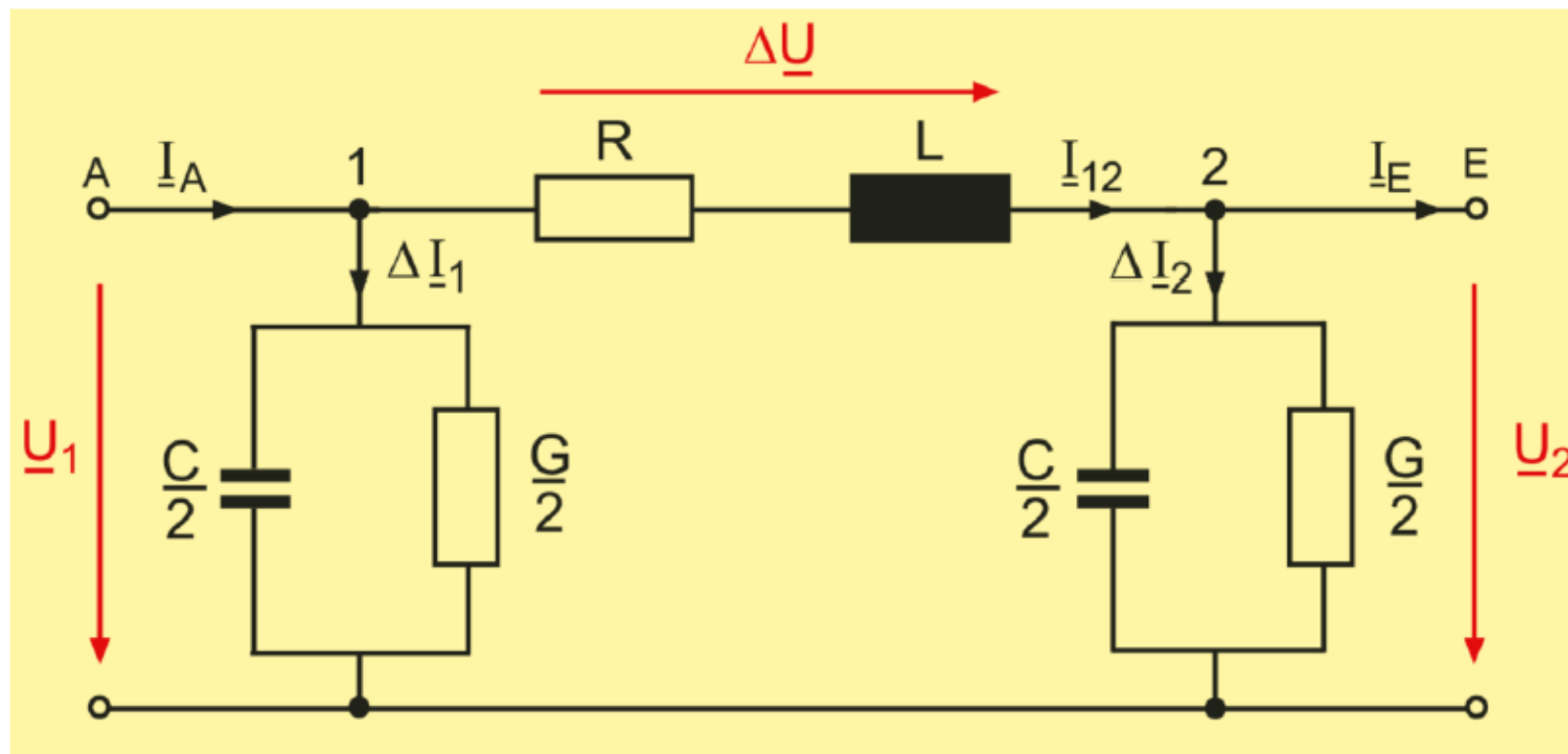
Equivalent Circuit of an short electrical line



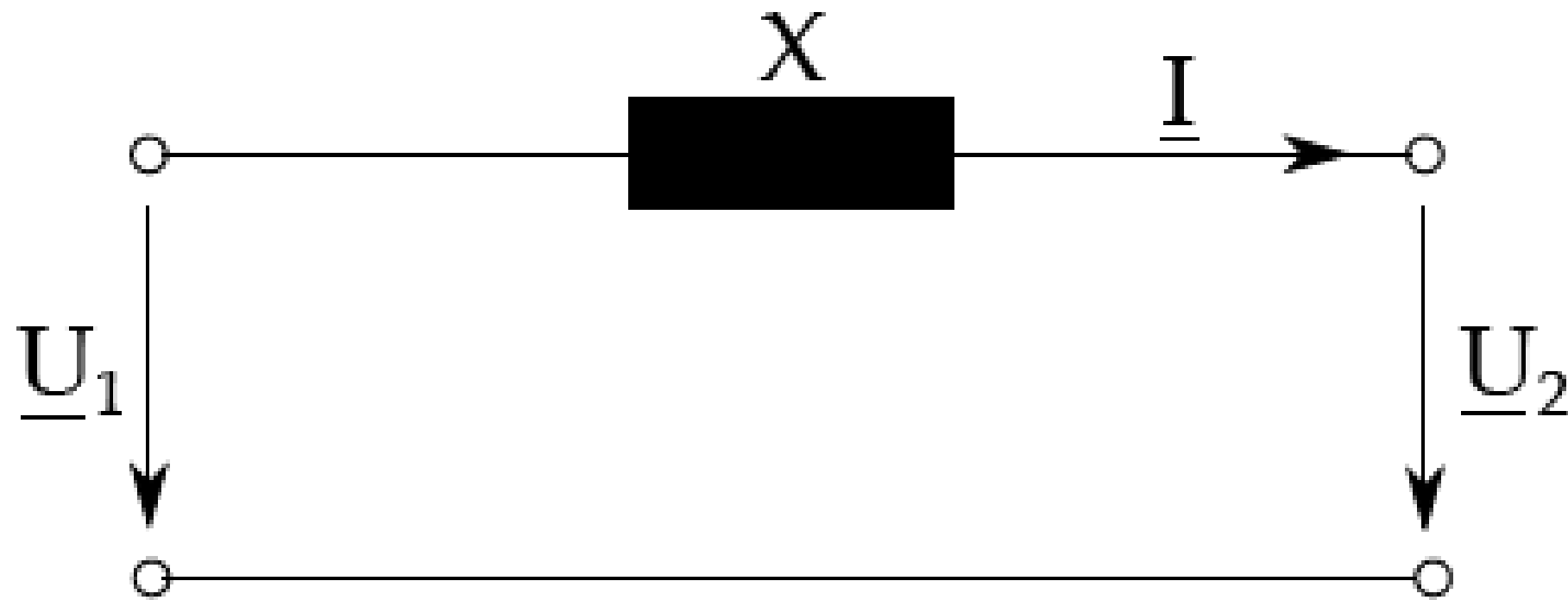
- ◆ Assumptions:
- ◆ Short overhead line < 100 km, short cable < 50 km
- ◆ Three phase symmetrical system

Reduce parameter

- ♦ $G = 0$ in low-, medium- and high-voltage grids
- ♦ $C = 0$ in overhead lines in low- and medium-voltage grids. However, in cables C has to be considered
- ♦ $L = 0$ in low-voltage grids. R is dominating in these grids
- ♦ $R = 0$ in high-voltage grids and L becomes dominating
- ♦ L and $C = 0$ in direct current grids. Only R becomes relevant



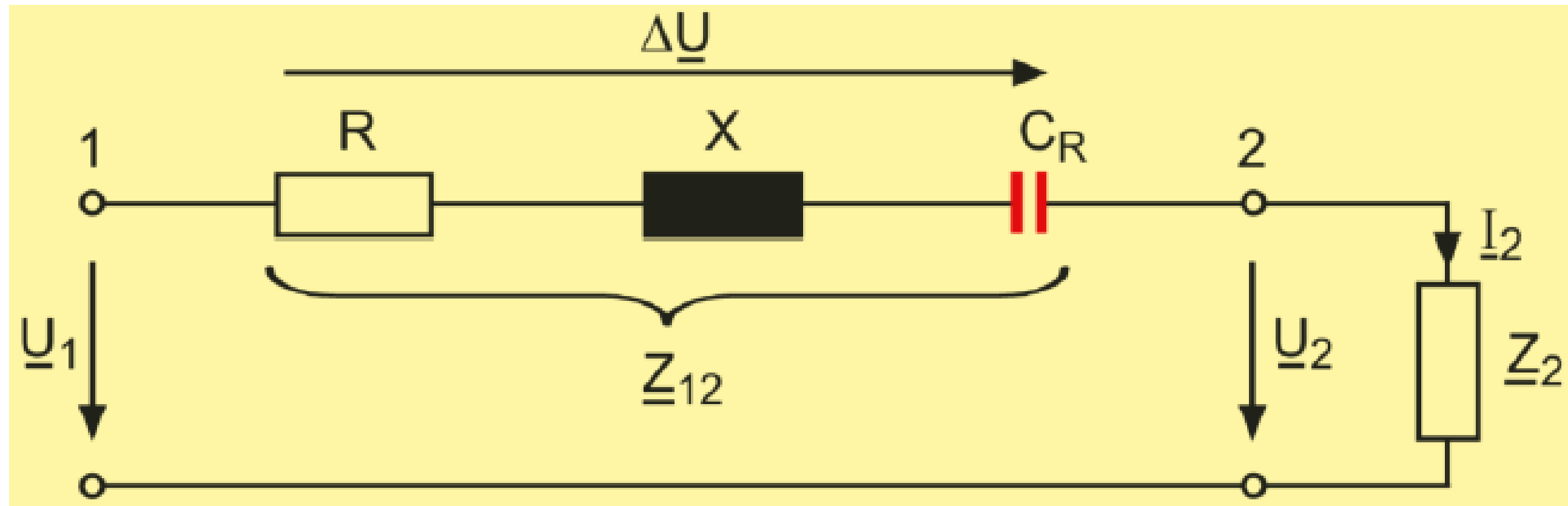
Transfer of Real Power



$$P = \frac{U_1 \cdot U_2}{X} \cdot \sin(\vartheta)$$

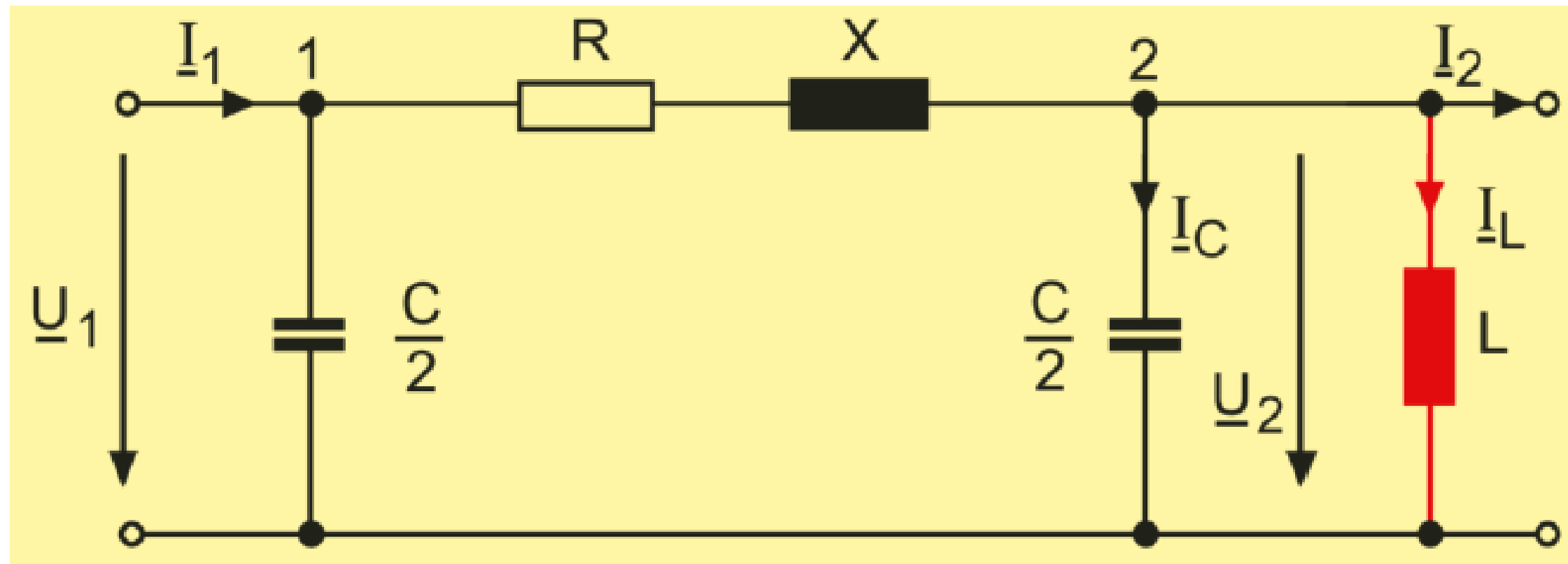
- ◆ Equivalent Circuit of an high voltage overhead line
- ◆ Theta = line angle ($\rho_1 - \rho_2$) $< 90^\circ$ for stability reason
- ◆ Theta usually below 45°

Serial compensation



- ♦ Capacitor is added in serial
- ♦ Increases voltage $\underline{U}_2$
- ♦ Capacitor banks are connected

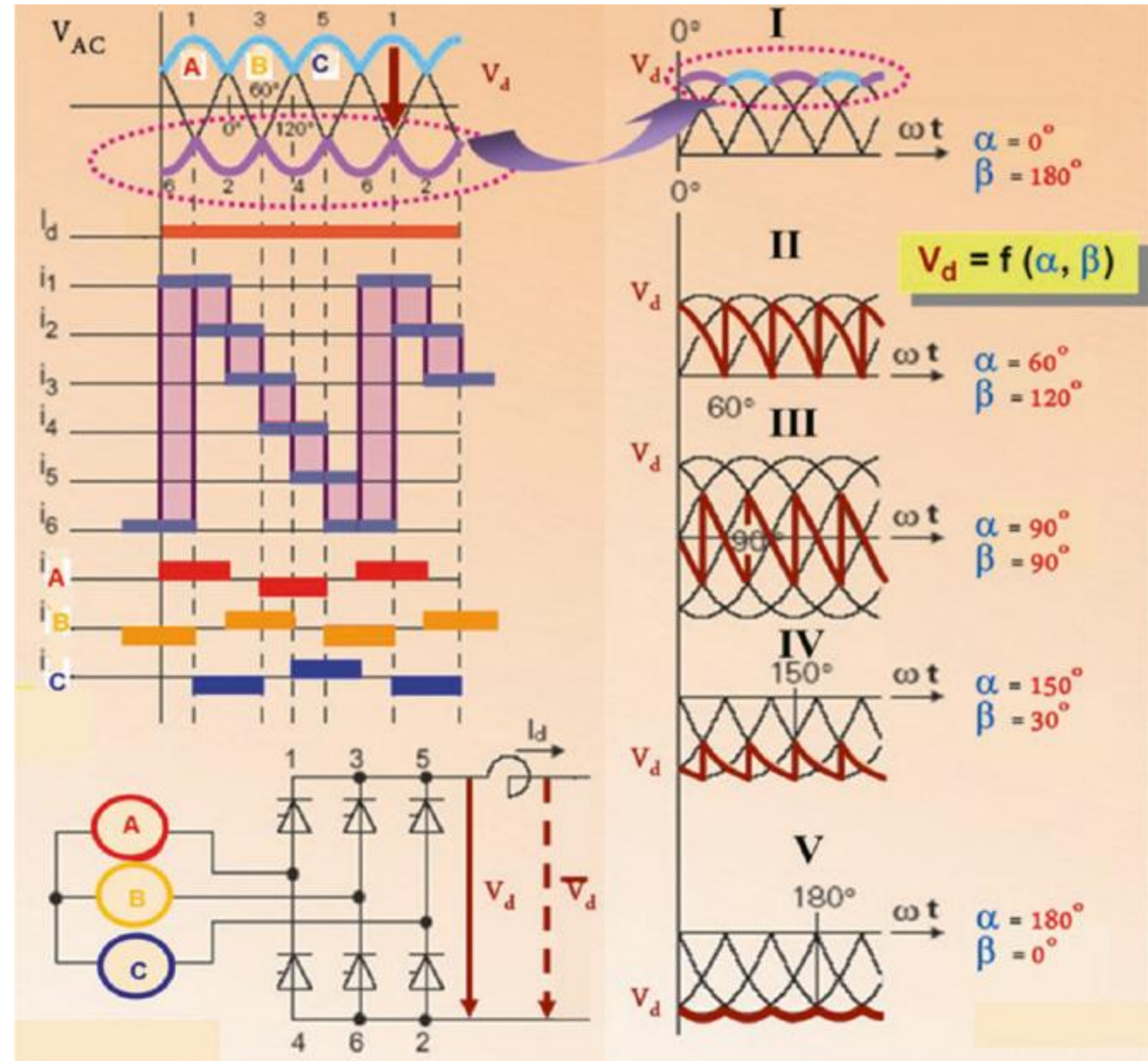
Parallel compensation



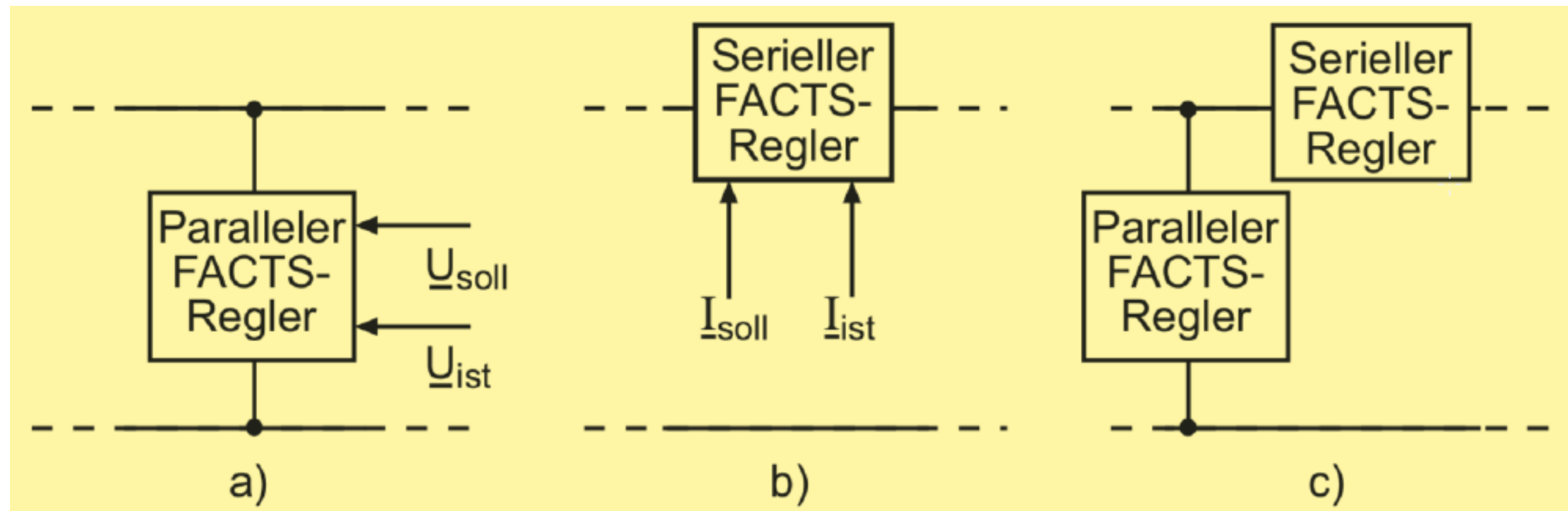
- ◆ Inductance parallel to the load
- ◆ Handles overvoltages and instability situations.
- ◆ E.g. during low load situations the capacitor causes an increase in voltage and line angle

Flexible AC Transmission Systems (FACTS)

- ◆ **Power electronics** are used together with reactors and capacitors to dynamically control power flow

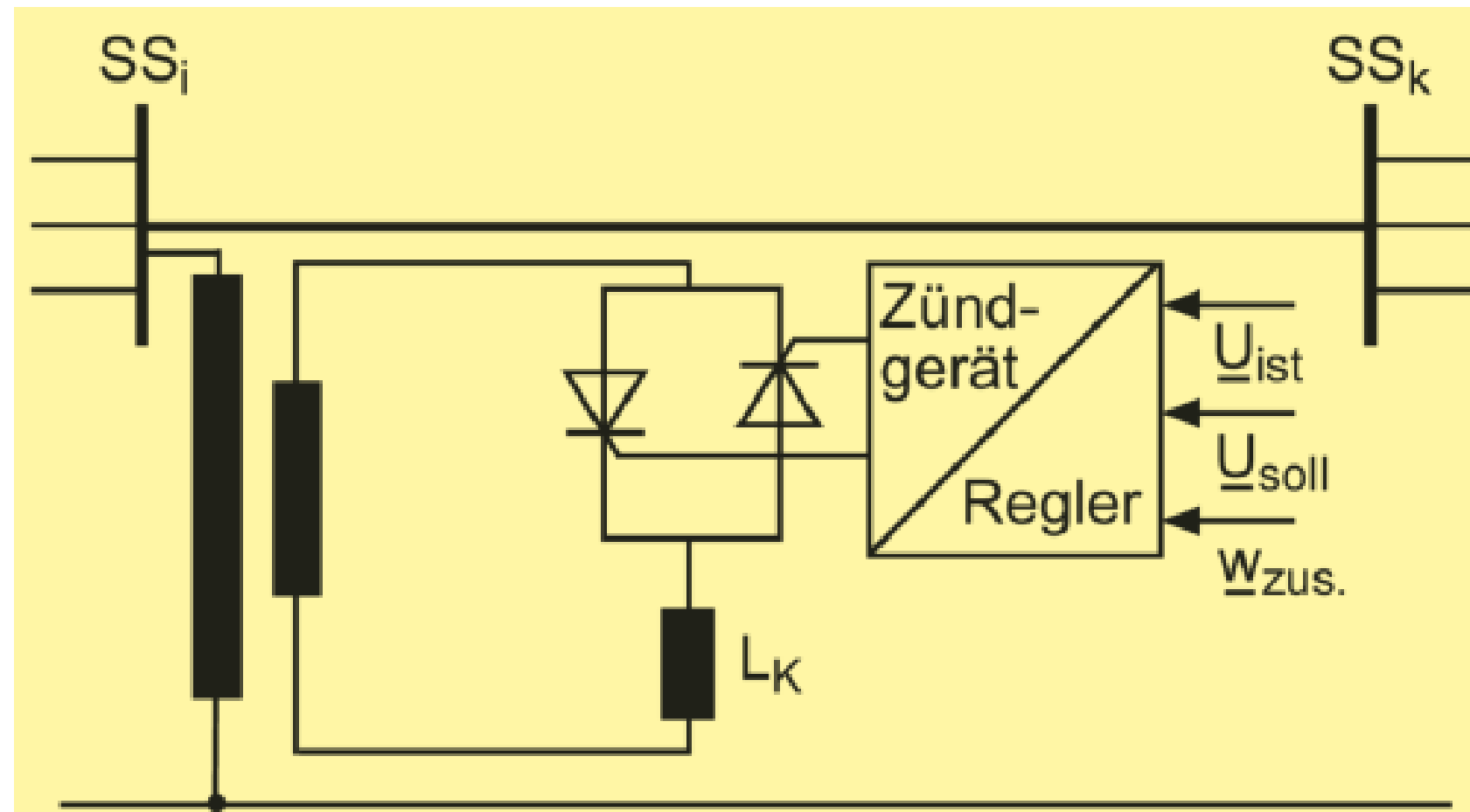


FACTS controller classification



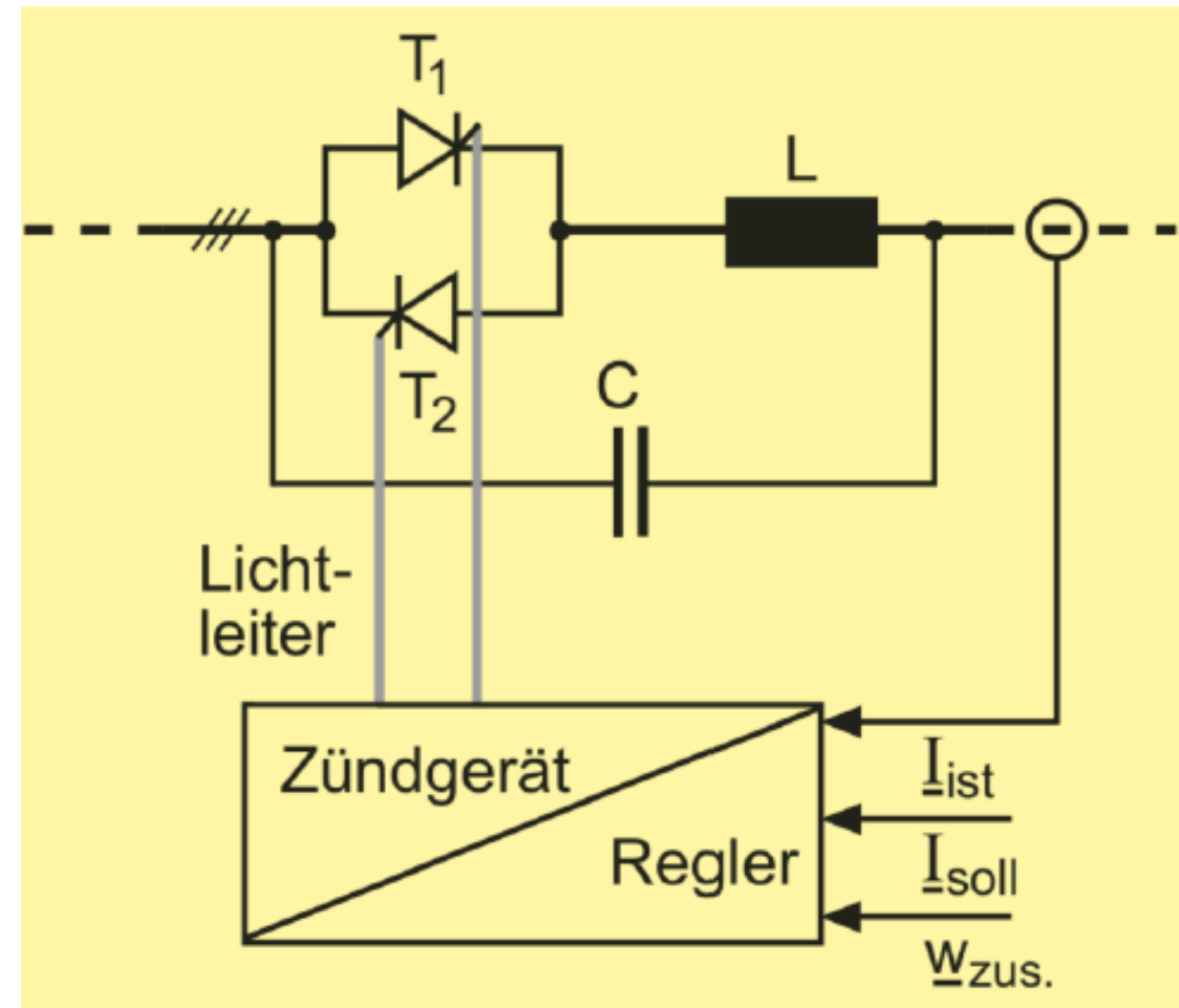
- a) Parallel FACTS Controller (Voltage Control)
- b) Serial FACTS Controller (Current Control)
- c) Combined FACTS Controller (With DC-Link)

Thyristor Controller Reactor (Parallel Example)



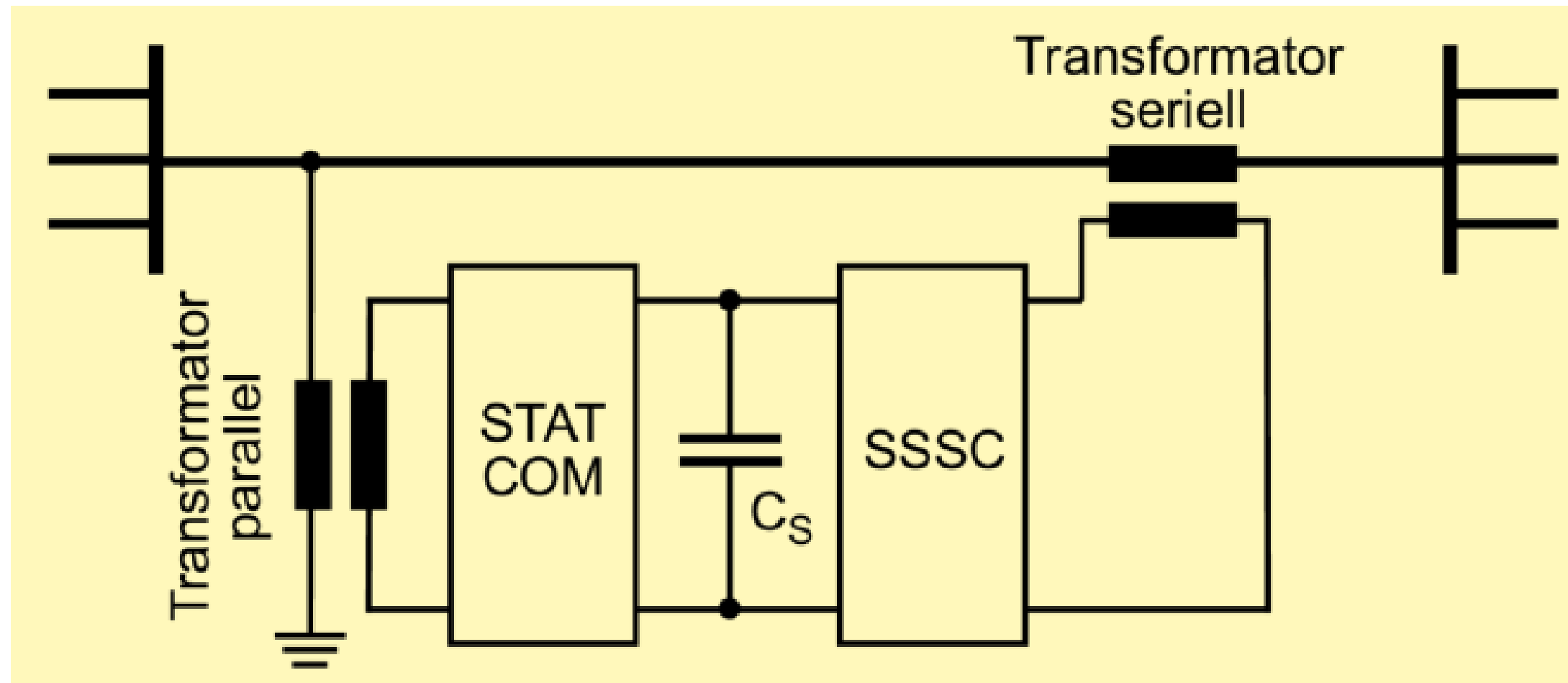
- ♦ Valves are switched by ignition device (dt. Zündgerät)
- ♦ It sequence comes from the controller (dt. Regler)

Thyristor Controller Series Capacitor (Serial Example)



- ◆ Valves control current flow through choke
- ◆ This controls the capacitor:
 - Valves on: Capacitor has no effect
 - Valves off: Capacitor is integrated into the circuit

Unified Power Flow Controller (UPFC)



- ♦ Combination of serial FACTs and parallel FACTs
- ♦ Connected via DC-Link and a Capacitor (C_s)

Electrical Energy Storage Systems (EESS)

- ◆ Classification is done according
 - Capacity
 - Time to make energy available

- ◆ Applications
 - Power Quality Improvements
 - Uninterrupted Power Supply
 - Energy Management

Power Quality Improvements

- ◆ Prevent
 - voltage sags
 - flicker
- ◆ Time range: seconds to a few minutes
- ◆ Technologies
 - Fly wheels
 - Magnetic energy storage
 - Super capacitors
 - Different types of batteries

Uninterrupted Power Supply

- ◆ Bridge time between energy loss until backup generator is ready
- ◆ Typical time: within minutes
- ◆ Technologies
 - Super capacitors
 - Different types of batteries

Energy Management

- ◆ Energy Storage for
 - peak shaving
 - power balancing
- ◆ Refinement
 - store during low price periods
 - discharge during high price periods
- ◆ Became more important with increasing share of fluctuating power producers
- ◆ Technologies
 - Pump Storage Power Plants
 - Compressed Air Energy Storage (CAES)
 - High energy batteries
 - Power to gas



www.fhv.at

Andreas Nenning
Winter Term 2026
Course Number 074742056202